# Comparing PID and Reinforcement-Learning (PPO) Controllers for Spacecraft Attitude Control

*A simulation study*

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Code and data: github.com/yasinkh81b-ui/ppo-vs-pid-spacecraft-attitude

**Abstract.** Reinforcement learning (RL) is increasingly proposed for spacecraft attitude control, but claims of superiority are often made against weakly tuned classical baselines. This study compares a PD controller, a PID controller with anti-windup, and two Proximal Policy Optimization (PPO) agents on a simulated rigid spacecraft with actuator saturation and an unknown constant disturbance torque. Classical gains were tuned by grid search on a held-out set of scenarios, and all controllers were tested on 100 unseen random scenarios. The tuned PID achieved the lowest steady-state error (median 0.007°) and settled within 1° in 3.6 s. The standard PPO agent settled in 4.4 s but kept a 0.42° residual error, because its observation contains no memory of past error. Adding the integral of the error to the observation reduced this residual about twelve-fold (0.035°) but slowed settling to 8.0 s. Both PPO agents used roughly a quarter less control torque than PD/PID. The study also documents how an early single-scenario comparison against a lightly tuned PID wrongly favored PPO, illustrating the importance of strong baselines.

## 1. Introduction

Attitude control keeps a spacecraft pointed in a desired direction despite disturbances such as solar radiation pressure, magnetic torques, and residual momentum. Quaternion-feedback PD/PID controllers are the long-standing baseline for this task [4, 5]. Recently, deep RL methods such as PPO [1] have been proposed as alternatives that learn a control policy from interaction rather than from an analytical design. A fair assessment of such methods requires a carefully tuned classical baseline, a held-out evaluation set, and metrics beyond a single trajectory.

This work asks a simple question: *in a controlled simulation, does a PPO policy outperform a properly tuned PID controller for rest-to-rest attitude maneuvers under saturation and an unknown constant disturbance?* A secondary hypothesis was that a memoryless PPO policy cannot cancel a constant disturbance, and that giving it the integral of the attitude error, the same information a PID integrator uses, would close that gap. The contributions are:

- a reproducible comparison protocol (separate tuning/training and test scenarios, four controllers, four metrics);
- an ablation testing whether an error-integral observation gives PPO the missing “memory”;
- open-source code, trained models and raw results, so that every number in this report can be regenerated.

## 2. Methods

### 2.1 Spacecraft model

The spacecraft is a rigid body with state $x = [q, \omega]$, where $q$ is the unit attitude quaternion (scalar first) and $\omega$ the body-frame angular velocity. The dynamics are

$$\mathrm{d}q/\mathrm{d}t = \tfrac{1}{2}\, q \otimes [0, \omega], \quad J\,\mathrm{d}\omega/\mathrm{d}t = \tau + d - \omega \times (J\omega),$$

with inertia $J = \mathrm{diag}(0.05, 0.06, 0.04)$ kg·m², commanded torque $\tau$ limited to ±0.05 N·m per axis, and an unknown constant disturbance torque $d$. Equations are integrated with a fourth-order Runge–Kutta scheme (step 0.02 s, quaternion renormalized every step); the controller acts every 0.1 s with a zero-order hold. The simulator was verified against two analytic checks: with zero torque, the angular momentum norm is conserved to about $10^{-17}$, and with a constant torque about one axis from rest, the angular velocity matches $\tau/J \cdot t$ to machine precision.

### 2.2 PD and PID controllers

Let $q_e = q_\mathrm{d}^{-1} \otimes q$ be the error quaternion and $e$ its vector part, multiplied by the sign of its scalar part so that the shorter rotation is always chosen. The control law is

$$\tau = -K_p\, e - K_d\, \omega - K_i \int e \, \mathrm{d}t,$$

followed by saturation. The PD controller is the case $K_i = 0$. To avoid integral windup during large slews, the integrator only accumulates when $\|e\| < 0.1$ (an error angle below about 11.5°) and the commanded torque is not saturated, and its state is clamped [5]. Gains were chosen by grid search over $K_p \in \{0.2, 0.4, 0.8, 1.6, 3.2\}$, $K_d \in \{0.1, 0.2, 0.4\}$ and, for the PID, $K_i \in \{0.01, 0.03, 0.06\}$, minimizing the mean of (settling time + 20 × final error) on 10 training scenarios. The selected gains are PD: $(K_p, K_d) = (1.6, 0.4)$ and PID: $(K_p, K_d, K_i) = (0.8, 0.2, 0.06)$.

### 2.3 PPO agents

The simulator was wrapped as a Gymnasium [3] environment. The observation is the error vector $e$ and the scaled angular velocity (6 values). The action is a 3-axis torque in $[-1, 1]$, scaled to the 0.05 N·m limit. The reward is dense:

$$r = -(\|e\| + 0.2\|\omega\| + 0.05\|u\|^2) + 0.1 \cdot [\|e\| < 0.0087],$$

where the bracket rewards being within about 1° of the target. Each 40 s episode starts from a random attitude (rotation of 20°–150° about a random axis), a random initial spin (±0.05 rad/s per axis) and a random constant disturbance (±0.003 N·m per axis); the target attitude is fixed. The **PPO+I** variant adds three observations, the scaled integral of $e$, accumulated under the same rule and clamp as the PID integrator; nothing else differs, so the comparison PPO vs. PPO+I isolates the effect of this memory. Both agents use Stable-Baselines3 [2] PPO with the default MLP policy, learning rate $3\times10^{-4}$, 2048 steps per update, batch size 64, discount 0.99, 300,000 environment steps (about 20 minutes on a laptop CPU), seed 0, and are evaluated deterministically.

### 2.4 Evaluation protocol

All four controllers use the same simulator, torque limit and control period. Tuning used scenario seeds 0–9; testing used 100 different scenarios (seeds 1000–1099) drawn from the same distribution as the RL training episodes, each simulated for 60 s. Metrics: *settling time* (last instant the attitude error exceeds 1°), *final error* (mean attitude error over the last 5 s), *RMS torque* (√ of the time-mean of $\|\tau\|^2$) and *torque jerk* (mean $\|\tau_k - \tau_{k-1}\|$ per control step).

## 3. Results

Table 1 summarizes the four controllers over the 100 test scenarios, and Figure 1 shows the error trajectories and the distribution of the metrics.

**Classical controllers.** PD and PID have nearly identical settling times (3.5 s and 3.6 s), but only the PID removes the effect of the constant disturbance: its final error (0.007°) is about 30 times smaller than the PD's (0.205°). **Standard PPO** settles slightly later (4.4 s) and plateaus at 0.416°, about twice the PD's residual error and 60 times the PID's. **PPO+I** reduces the residual error about twelve-fold relative to PPO (0.416° → 0.035°), confirming

that the missing memory was a major cause of the bias, but it remains about five times worse than the PID and its median settling time nearly doubles (8.0 s), because the error decays slowly in the 1° to 0.1° range (Figure 1, left). **Control effort.** Both RL agents use about 23–28% less RMS torque than PD/PID and produce slightly smoother commands. All four controllers brought the attitude error below 1° in all 100 scenarios.

| Controller | Settling time to 1° median (90th pct) | Final error (median) | RMS torque (N·m) | Torque jerk | Settled <1° |
|---|---|---|---|---|---|
| PD (tuned) | 3.5 s (4.9 s) | 0.205° | 0.0143 | 0.0007 | 100% |
| PID, anti-windup (tuned) | 3.6 s (5.4 s) | **0.007°** | 0.0139 | 0.0006 | 100% |
| PPO | 4.4 s (5.7 s) | 0.416° | 0.0107 | 0.0005 | 100% |
| PPO+I | 8.0 s (10.5 s) | 0.035° | **0.0103** | **0.0004** | 100% |

*Table 1. Results over 100 unseen random scenarios (60 s each, control period 0.1 s, torque limit 0.05 N·m per axis). Medians are reported unless noted. Best value in bold for final error, RMS torque and jerk; PD and PID have nearly identical settling times.*

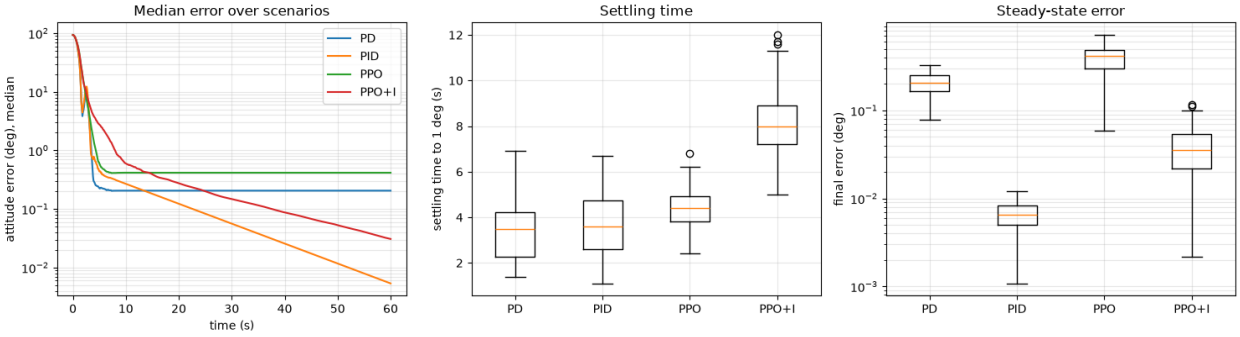



*Figure 1. Left: median attitude error over the 100 scenarios (log scale). Centre: distribution of settling times. Right: distribution of final errors. PD and PPO plateau at a nonzero error under the constant disturbance, the PID error keeps decreasing, and PPO+I decreases slowly.*

**Lesson on baselines.** An early test on a single 90° slew, using a lightly hand-tuned PID, suggested that PPO was about five times faster (4.1 s vs. 20.4 s to reach 1°). Systematic tuning of the classical controllers and testing on 100 scenarios reversed this conclusion. This is why the protocol above separates tuning from testing.

## 4. Discussion

**Why the PID wins here.** The task is a smooth, nearly linear regulation problem for which a feedback law with an integrator is close to ideal; the constant disturbance is exactly what the integral term is designed to reject. A memoryless policy cannot infer an unknown constant torque from a single observation, which explains PPO's steady offset. Giving the policy the error integral provided the necessary information, and the offset dropped accordingly, so the information was used.

**Why PPO+I is slow.** The cause was not isolated in this study. Plausible explanations are the reward design (a linear error penalty with a nonzero effort penalty gives a weak incentive to correct small errors quickly), the limited training budget (300k steps), and training variance (one seed per variant). Each is testable, and none is claimed here.

**Control effort.** The roughly 25% lower torque of the RL agents should not be read as an intrinsic advantage: their reward explicitly penalizes effort, while the PD/PID gains were tuned only for settling time and final error. A fair multi-objective comparison would add an effort term to the classical tuning objective or compare Pareto fronts.

**Limitations.**

- Simulation only: no sensor noise, actuator dynamics, flexible modes or delays; the inertia is known and fixed.
- Only constant disturbances were considered; the test distribution matches the RL training distribution, so generalization to other conditions was not assessed.
- One training run per RL variant, so seed-to-seed variance is unknown, and no formal significance tests were performed (medians and percentiles over 100 scenarios are reported).
- The classical gains come from a small grid search; finer or multi-objective tuning could shift the classical results.

## 5. Future Work

- Train several seeds per variant and report confidence intervals; tune the reward (e.g., a stronger incentive for small errors) and test longer training.
- Evaluate robustness: sensor noise, time-varying disturbances, inertia uncertainty and reaction-wheel dynamics; add domain randomization during training.
- Compare against stronger classical baselines (LQR, model-predictive control) and use multi-objective (effort vs. accuracy) comparisons.

## 6. Conclusion

In this simulation study, a properly tuned PID controller outperformed PPO in settling speed and steady-state accuracy under saturation and a constant disturbance, while the PPO agents achieved lower control effort. Adding the error integral to the PPO observation removed most of its steady-state bias but did not match the PID. The more general result is methodological: comparisons between RL and classical controllers should be made against carefully tuned baselines and on held-out scenarios. All code, trained models and raw results are available in the open-source repository at github.com/yasinkh81b-ui/ppo-vs-pid-spacecraft-attitude.

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## References

[1] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, O. Klimov. Proximal Policy Optimization Algorithms. arXiv:1707.06347, 2017.

[2] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, N. Dormann. Stable-Baselines3: Reliable Reinforcement Learning Implementations. Journal of Machine Learning Research 22(268):1–8, 2021.

[3] M. Towers et al. Gymnasium: A Standard Interface for Reinforcement Learning Environments. arXiv:2407.17032, 2024.

[4] B. Wie, P. M. Barba. Quaternion Feedback for Spacecraft Large Angle Maneuvers. Journal of Guidance, Control, and Dynamics 8(3):360–365, 1985.

[5] K. J. Åström, T. Hägglund. Advanced PID Control. ISA, 2006.